

Syed Ali Abbas Rahil

B.Tech — Computer Science and Engineering
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Patna	8.44/10	2020–2024
Senior Secondary	CISCE Board	98.40%	2020
Secondary	ICSE Board	97.67%	2018

EXPERIENCE

- OTSUKA CORPORATION** Tokyo, Japan
AI Engineer (Full-Time) October 2024 – Present
 - Engineered agentic **RAG chatbots** for **1,000+** monthly users using ELK stack, LangChain, and Knowledge Graphs with long-term memory; improved retrieval accuracy by **31%** and reduced search latency by **23%**.
 - Built an enterprise **Knowledge Management System** with a 2-stage LLM evaluation pipeline that auto-generates QA pairs from **500+** internal documents; increased knowledge accessibility by **13%** for **1,000+** sales representatives.
 - Deployed **Bayesian leaderboards** and multi-tenant **RBAC** across **3** data sources (Microsoft Teams, SharePoint, CSV); secured role-based access for **5+** internal teams.
 - Fine-tuned **3+** Large Language Models via **SFT and RLHF** on multi-node on-premise GPU clusters using LoRA/QLoRA; achieved **15%** improvement on Question Answering benchmarks.
- OTSUKA CORPORATION** Tokyo, Japan
AI Research Intern June 2023 – July 2023
 - Developed **multi-agent LLM simulations** with LangChain in a **2-month** research engagement; reduced task completion time by **15%** in automated workflow pipelines.
 - Prototyped **3** autonomous agents with **adaptive memory** and goal-planning; delivered end-to-end multi-turn collaborative reasoning across **3+** complex task scenarios.

PROJECTS

- AGENT FORGE – MULTI-TOOL AI AGENT PLATFORM** March 2025
Personal Project GitHub
 - Created a full-stack multi-agent chat platform (**FastAPI + React/TypeScript**) with **2** configurable agent personas and **4** integrated tools; enabled real-time web search, Wikipedia lookup, and persistent chat history via LangChain tool-calling.
 - Implemented LLM-powered **long-term memory** that auto-extracts user preferences from conversations using Pydantic structured outputs; persisted chat history and user profiles in **SQLite** with full session CRUD operations.
- RAG BENCHMARK TOOLKIT** December 2024
Personal Project GitHub
 - Benchmarked **100+** RAG pipeline configurations across **4** document processors, **9** chunking strategies, and **4** search methods (BM25, KNN, Hybrid) on Elasticsearch 8.x; integrated cross-encoder reranking to optimize retrieval quality.
 - Automated end-to-end evaluation using **MRR, RAGAS Context Precision, ROUGE, and semantic similarity** across **3** LLM families (LLaMA, GPT, Qwen); replaced manual benchmarking with a single YAML-driven test pipeline.

TECHNICAL SKILLS

- Programming:** Python, C, C++, React/Next.js, SQL, Data Structures & Algorithms, Probability & Statistics
- ML/AI:** PyTorch, TensorFlow, Axolotl, DeepEval, MCP, LangChain, CrewAI, Mixture of Experts (MoE)
- Databases:** Elasticsearch, MySQL, MongoDB, PostgreSQL, Neo4j
- MLOps:** MLflow, LangFuse, AWS/Azure, Redis, vLLM, Docker, Kubernetes

ACHIEVEMENTS & LEADERSHIP

- Secured **AIR 2796** in JEE Advanced 2020 (top 0.3% of 150,000+ candidates nationwide).
- Solved **600+** **DSA problems** across Codeforces (Specialist), CodeChef (1992), and LeetCode.
- Published **DisT5: A Text-to-Text Transformer Model for Disaster Events** in *Online Social Networks and Media (Elsevier)*, 2026.
- Completed **Python for Machine Learning & Data Science** certification on Udemy.
- Achieved **Japanese Language Proficiency Test (NAT) N5** qualification.
- Coordinator – Creative Writing & Social Media, E-Cell, IIT Patna.